

Assessing Professor Effectiveness (APE)

Introduction to Data Science (DS-GA 1001) Capstone Project

Group 11

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0. Data Cleaning and Preprocessing

Prior to analysis, several data cleaning and preprocessing steps were performed to ensure consistency across numerical, categorical, and tag-based variables.

Three datasets (`rmpCapstoneQual.csv`, `rmpCapstoneNum.csv`, and `rmpCapstoneTags.csv`) were merged column-wise to create a unified professor-level dataset. Initial inspection revealed 19,889 rows missing simultaneously across key qualitative and numerical variables, including major, university, state, average rating, difficulty, rating counts, and pepper indicators. These rows also contained no teaching tag information and were removed, as they provided no usable information.

Observations with fewer than four ratings were excluded, as such small sample sizes were considered insufficient to produce stable estimates of average ratings and more likely to yield extreme or noisy values. This filtering reduced the dataset from approximately 70,000 observations to 31,951 rows.

Gender information was originally provided through separate binary indicators for male and female, which resulted in ambiguous cases where both indicators were zero or both were one. A single categorical gender variable with three levels (male, female, unknown) was constructed to resolve this ambiguity.

Tags consisted of 20 indicators describing instructional characteristics. In the raw data, extreme imbalance was observed within individual rows, where certain tags appeared hundreds of times while others appeared only once for the same professor. Using raw tag counts under such conditions would cause a small number of high-frequency tags to dominate the analysis. To address this issue, all tag values were normalized by the total number of tags per professor, allowing the analysis to reflect relative tag composition rather than absolute frequency.

All analyses involving randomization (e.g., resampling or data splitting) were made reproducible by fixing the random number generator seed to a constant value derived from Yoon Seol's NYU N-number prior to execution.

1. Gender Bias in Average Ratings

We tested whether there is evidence of a pro-male gender bias in student evaluations by examining whether male professors tend to receive higher ratings than female professors. Because student ratings are psychological and ordinal in nature and may not satisfy normality assumptions, we used a one-sided Mann-Whitney U test, which compares whether ratings for one group tend to be higher than those for another without assuming a specific distribution. The alternative hypothesis was that ratings for male professors are stochastically larger than those for female professors. Statistical significance was assessed at $\alpha = 0.005$, as specified in the assignment.

The descriptive results show a small difference in central tendency: the median average rating was 4.20 for male professors and 4.10 for female professors. The Mann-Whitney U test yielded a test statistic of $U = 69,429,526$ and a one-sided p -value of 3.93×10^{-5} , which is well below the significance threshold. We therefore reject the null hypothesis and find statistically significant evidence consistent with a pro-male gender bias in student ratings in this dataset.

However, the magnitude of the difference is modest. The distributions of ratings for male and female professors overlap substantially, and the observed shift corresponds to only a 0.1 difference in median rating on a 1–5 scale. Thus, while the effect is statistically detectable, it represents a small practical difference rather than a large disparity in evaluations.

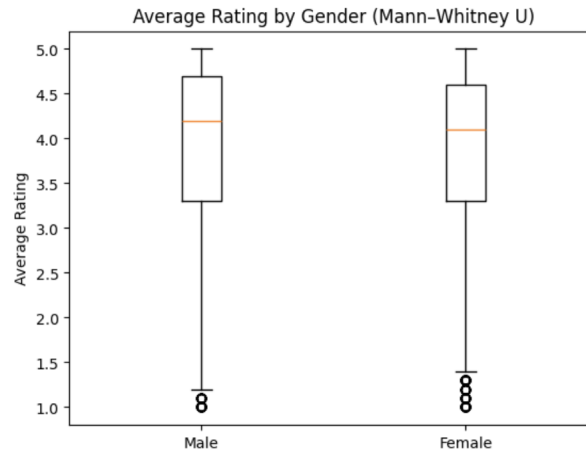


Figure 1: Distribution of average ratings by gender.

2. Gender Differences in Rating Dispersion

To examine whether there is a gender difference in the spread (dispersion) of student ratings, we focused on differences in variability rather than central tendency. Because rating distributions are bounded, skewed, and ordinal in nature, we used the median absolute deviation (MAD) as a robust measure of dispersion instead of variance or standard deviation. MAD captures how far ratings typically deviate from the median and is less sensitive to extreme values, making it well suited for student evaluation data.

To assess statistical significance, we conducted a two-sided permutation test using the difference in MAD between male and female professors (Male – Female) as the test statistic. Under the null hypothesis that rating dispersion is identical across genders, professor gender labels were randomly permuted while preserving the original sample sizes. We generated a null distribution using 10,000 permutations and evaluated statistical significance at $\alpha = 0.005$, as specified in the assignment.

The observed MAD difference was effectively zero, indicating nearly identical dispersion in ratings for male and female professors. The permutation test yielded a p -value of 0.5239, which is far above the significance threshold. We therefore fail to reject the null hypothesis and find no statistically significant evidence of a gender difference in rating dispersion as measured by MAD.

The permutation null distribution, shown in Figure reffig:q2, confirms this result: the observed MAD difference lies near the center of the distribution and well within the range expected under the null hypothesis. Overall, while earlier analyses detected small gender differences in average ratings, this analysis suggests that students do not differ in how consistently they rate male versus female professors, indicating similar levels of agreement or disagreement within each group.

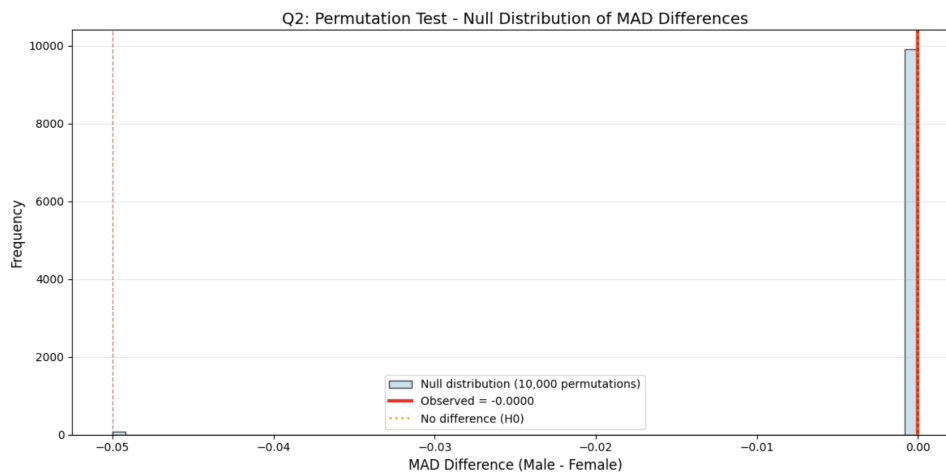


Figure 2: Permutation Test - Null Distribution of MAD Differences

3. Effect Size Estimation with Bootstrap Confidence Intervals

We estimated the likely size of both gender effects identified in Questions 1 and 2 by constructing 95% bootstrap confidence intervals. Because student ratings are ordinal and do not necessarily satisfy parametric assumptions, we summarized the difference in average ratings using the median difference and the difference in dispersion using the difference in standard deviation.

For the gender difference in average rating, the estimated median difference (Male – Female) was 0.10 rating points. The 95% bootstrap confidence interval was [0.10, 0.10], indicating a highly stable estimate in this dataset. This suggests that male professors are typically rated about one tenth of a point higher than female professors on the 1–5 rating scale. While statistically precise, this difference is small in practical terms.

For the gender difference in rating dispersion, we estimated the effect as the difference in standard deviation (Male – Female). The observed difference was -0.044 , with a 95% bootstrap confidence interval of $[-0.061, -0.027]$. This indicates that ratings for female professors are modestly more variable than those for male professors. Although the difference is not large, the confidence interval excludes zero, suggesting a consistent spread difference in the population represented by this dataset.

Figures illustrate the bootstrap distributions for both effect sizes, with the observed statistics and confidence interval bounds marked. Overall, both effects appear small in magnitude but are precisely estimated due to the large sample size.

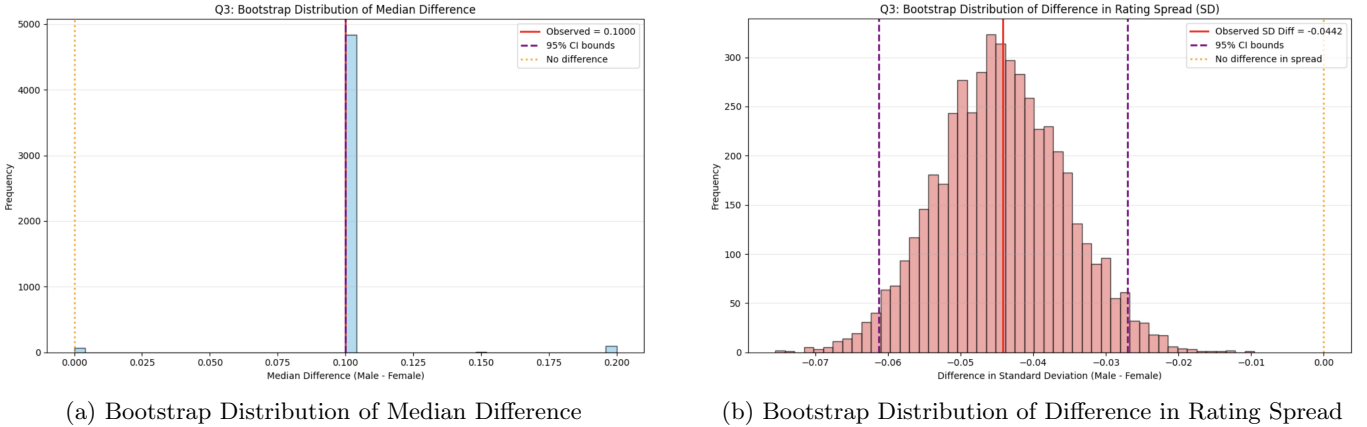


Figure 3: Effect size estimation with bootstrap distributions (Q3).

4. Gender Differences in Teaching Tags

We examined whether there were gender differences in the tags awarded by students. For each of the 20 tags, we compared the distribution of the normalized tag proportion between male and female professors using a two-sided Mann–Whitney U test. Because this analysis involves multiple simultaneous hypothesis tests, we controlled the family-wise error rate using a Bonferroni correction, setting the significance threshold to $\alpha_{\text{corr}} = 0.005/20 = 0.00025$.

The three most gendered tags were *hilarious* ($U = 81,954,877.5$, $p = 8.62 \times 10^{-218}$), *amazing lectures* ($U = 74,553,982.5$, $p = 9.05 \times 10^{-53}$), and *respected* ($U = 73,752,303.0$, $p = 1.25 \times 10^{-37}$). All remain statistically significant after Bonferroni correction.

In contrast, the least gendered tags were *don't skip class*, *tough grader*, and *pop quizzes*, none of which were significant under the adjusted threshold. Overall, gender differences in tagging behavior are concentrated in subjective, perception-based tags rather than course-structure indicators.

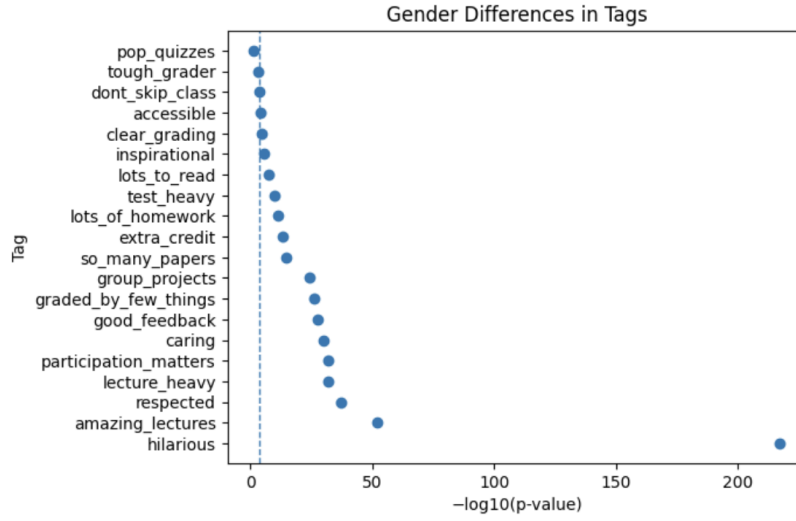


Figure 4: Gender Differences in Tags

5. Gender Differences in Average Difficulty

To investigate whether a gender difference exists in terms of average difficulty, we utilized the Mann–Whitney U test. We selected this non-parametric test because difficulty ratings are not normally distributed and because the psychological distance between scale values may be inconsistent; for example, students may perceive the gap between a difficulty of 2 and 3 differently from the gap between 4 and 5. By comparing the ranks of evaluations rather than raw means, the analysis remains robust to these ordinal characteristics.

Our analysis of the filtered dataset revealed that both male and female professors received an identical median difficulty rating of 2.9000. The two-sided Mann–Whitney U test yielded a test statistic of $U = 67,549,614.00$ with a p -value of 0.7883. Because $p > 0.005$, we fail to reject the null hypothesis and conclude that there is no statistically significant difference in average difficulty between genders.

This result is also confirmed visually by the boxplot, in which the medians, quartiles, and whiskers for both groups are nearly indistinguishable. Overall, students appear to judge the workload and rigor of male and female professors using comparable standards.

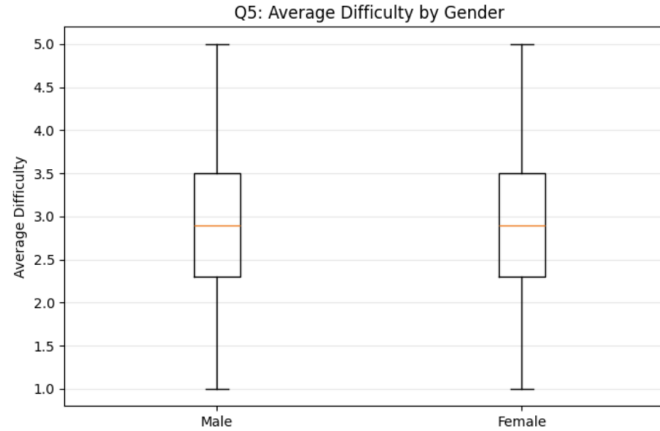


Figure 5: Distribution of average difficulty by gender.

6. Effect Size of Difficulty Difference

To quantify the likely size of the gender effect on perceived difficulty, we conducted a bootstrap analysis with 5,000 replications. Bootstrapping allows estimation of the sampling distribution of the median difference without assuming normality.

The observed difference in median difficulty (Male – Female) was 0.0000. The resulting 95% bootstrap confidence interval was $[-0.1000, 0.1000]$. Because this interval is centered at zero and spans both negative and positive values, it reinforces the conclusion from Question 5 that there is no meaningful gender difference in perceived difficulty.

The bootstrap distribution is highly concentrated around zero, indicating that any observed differences are well within the range of random sampling variation. These findings suggest that the likely size of any gender bias in difficulty is effectively zero.

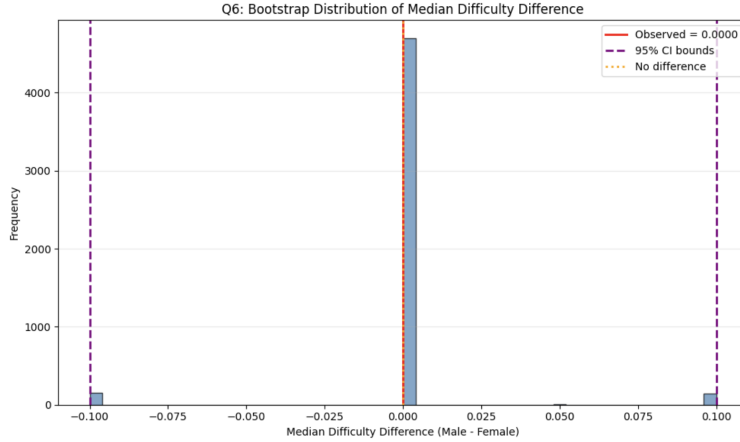


Figure 6: Bootstrap Distribution of Median Difficulty Difference

7. Predicting Average Rating from Numerical Variables

To predict professors’ average ratings using numerical variables from `rmpCapstoneNum.csv`, we constructed a regression model including all numerical predictors. Gender was incorporated using dummy encoding, with male professors treated as the reference category. This approach avoids imposing artificial ordering or distance between gender categories, which would be implied by integer encoding.

Several predictors exhibited moderate correlations, including a negative correlation between average difficulty and the proportion of students willing to take the course again ($r = -0.52$), and a positive correlation between pepper indicators and take-again percentage ($r = 0.44$). To address potential multicollinearity, Ridge regression was employed.

The fitted model achieved strong predictive performance, with an R^2 of 0.804 and a root mean squared error (RMSE) of 0.374. These results indicate that numerical features explain a substantial proportion of the variance in average ratings.

A coefficient plot of standardized Ridge estimates shows that `take_again_percent` is the most strongly predictive variable, exhibiting a large positive association with ratings. Average difficulty shows a negative association, while pepper indicators contribute positively but with smaller effect sizes.

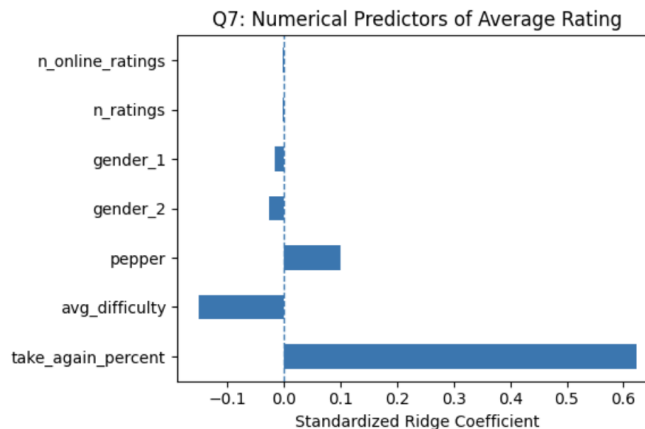


Figure 7: Numerical Predictors of Average Rating

8. Predicting Average Rating from Teaching Tags

To assess whether qualitative teaching descriptors provide additional explanatory power, we constructed a regression model predicting average ratings using the full set of normalized teaching tags.

Correlation analysis revealed dependence among several tags, particularly among positive descriptors such as *good feedback*, *amazing lectures*, *respected*, and *caring*, with pairwise correlations in the range of approximately $r = 0.25$ to 0.35 . This multicollinearity motivated the use of Ridge regression.

The fitted model achieved an R^2 of 0.683 and an RMSE of 0.545. While this performance is lower than that of the numerical model in Question 7, it demonstrates that teaching tags explain a meaningful portion of the variation in average ratings.

A coefficient plot of standardized Ridge estimates shows that *good feedback* is the most strongly predictive tag, followed by other positive teaching descriptors such as *amazing lectures* and *respected*. In contrast, *tough grader* exhibits a negative association with average rating, suggesting that stricter grading perceptions are associated with lower evaluations.

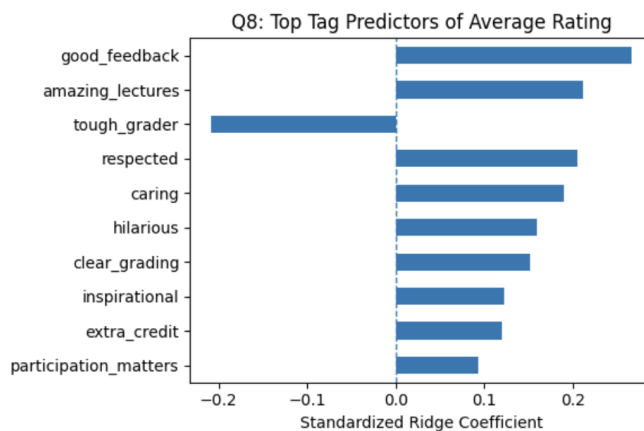


Figure 8: Top Tag Predictors of Average Rating

9. Predicting Average Difficulty from Teaching Tags

Finally, we constructed a regression model predicting professors' average difficulty using the same set of normalized teaching tags. As in Question 8, substantial correlations among tag variables motivated the use of Ridge regression to address multicollinearity.

The model achieved an R^2 of 0.520 and an RMSE of 0.572, indicating moderate predictive performance. This suggests that while teaching tags capture meaningful information about perceived course difficulty, they do not fully account for all factors influencing students' difficulty assessments.

A coefficient plot of standardized Ridge estimates shows that *tough grader* is the most strongly predictive tag, with a large positive association with average difficulty. Tags such as *test heavy* also contribute positively, while descriptors including *clear grading*, *caring*, and *hilarious* are negatively associated with difficulty. These results indicate that students use teaching tags consistently to describe both course rigor and instructor characteristics.

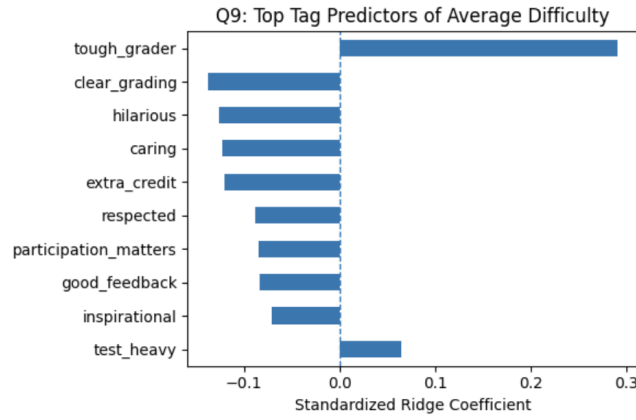


Figure 9: Top Tag Predictors of Average Difficulty

10. Predicting “Pepper” Status Using Classification

We developed a classification model to predict whether a professor receives a “pepper” using all available numerical features and normalized qualitative teaching tags. For this task, we employed Logistic Regression.

A critical consideration for this model was class imbalance. The ratio of “No Pepper” to “Pepper” professors is approximately 1.15:1. To address this issue, we used the `class_weight = balanced` option, which assigns weights inversely proportional to class frequencies. This ensures that the model does not become biased toward the majority class.

The model demonstrated strong predictive performance, achieving an area under the receiver operating characteristic curve (AUROC) of 0.8151. Given the imbalance considerations, we also emphasized the area under the precision–recall curve (AUPRC), which was 0.7781. This indicates strong performance in identifying “hot” professors in particular.

As illustrated by the ROC and precision–recall curves and the feature importance bar chart, the most strongly predictive factor was `avg_rating`, with a standardized coefficient of +1.1024. This was followed by `n_ratings` and `avg_difficulty`. In contrast, tags such as `tough grader` and `lots of homework` were among the strongest negative predictors.

Overall, these results suggest that students’ perceptions of a professor’s “hotness” are closely intertwined with overall teaching quality and platform visibility, rather than being driven by superficial or isolated teaching characteristics.

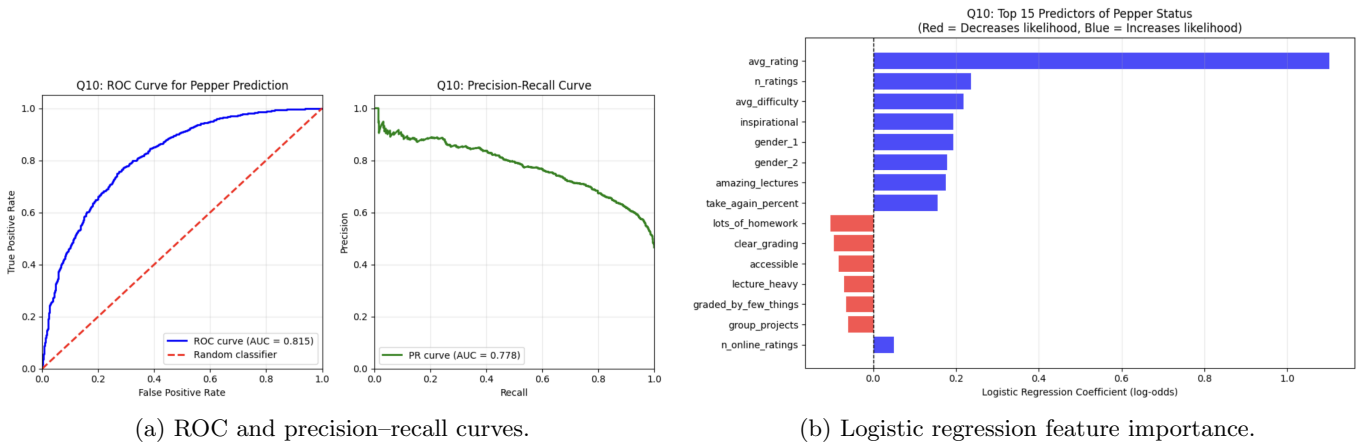


Figure 10: Classification results for predicting “pepper” status (Q10).

Extra Credit: Rating–Difficulty Relationship Across Majors

To investigate whether the relationship between course difficulty and professor ratings differs across academic fields, the data were grouped by major and analyzed separately. For each major with at least 200 observations, a simple linear regression was fit with average rating as the outcome and average difficulty as the predictor. The estimated

slope represents the expected change in rating associated with a one-unit increase in difficulty within that major.

Across all majors, higher course difficulty is associated with lower average ratings, but the magnitude of this relationship varies substantially by field. Majors such as Mathematics, Accounting, Biology, Chemistry, and Physics exhibit the steepest negative slopes, indicating that ratings decline sharply as difficulty increases. For example, in Mathematics, a one-point increase in average difficulty corresponds to nearly a one-point decrease in average rating.

In contrast, majors such as Art, Languages, Marketing, Political Science, and Spanish show much weaker negative slopes. In these fields, ratings remain relatively stable across difficulty levels, suggesting that students place less weight on course rigor when evaluating instructors.

Overall, this analysis demonstrates that the rating–difficulty tradeoff is not uniform across disciplines. Students’ responses to course difficulty depend strongly on academic context, highlighting a key limitation of directly comparing teaching evaluations across fields without accounting for disciplinary differences.

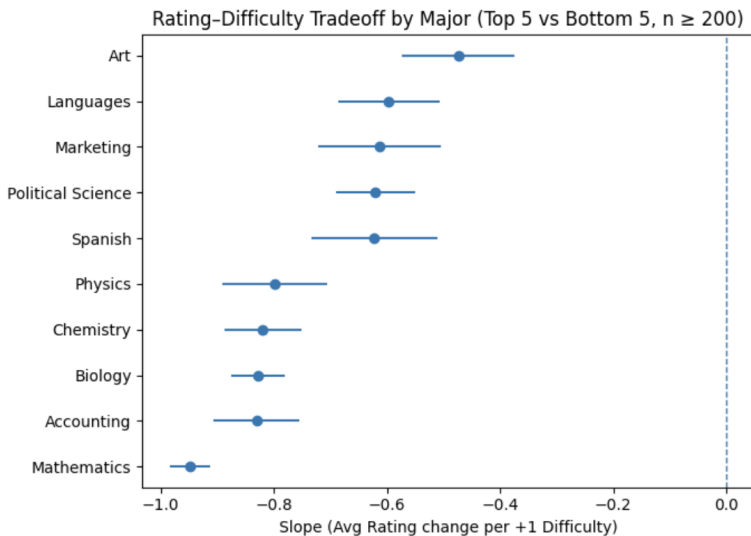


Figure 11: Distribution of average ratings by gender.